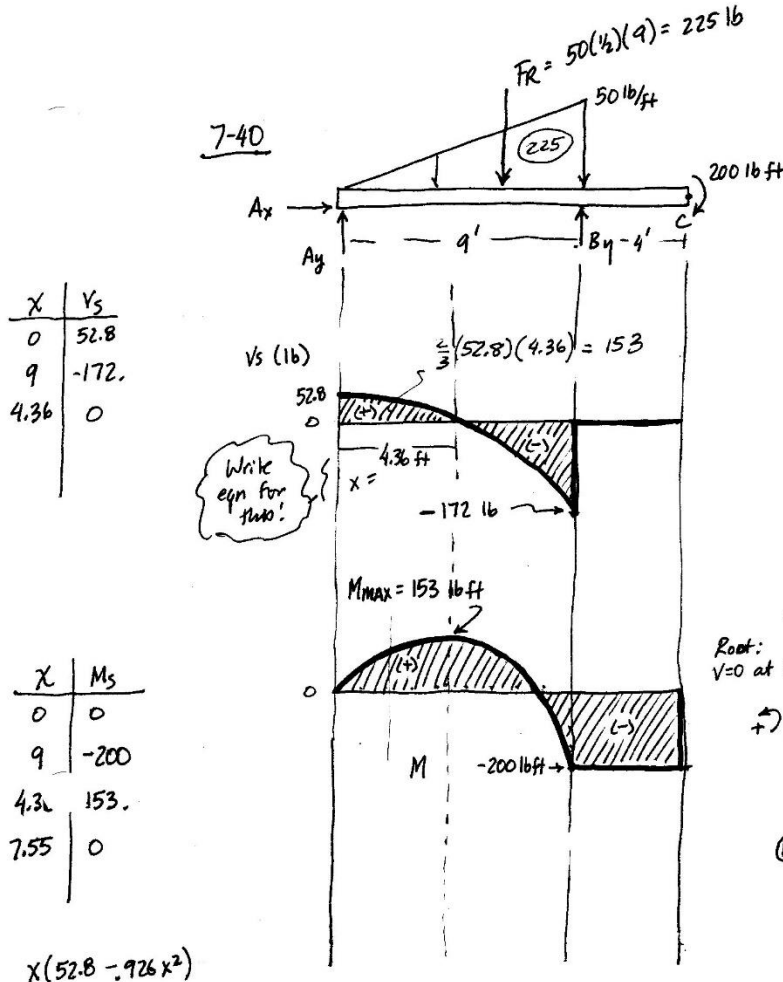


Statics Internal Forces: Shear and Bending Moment Diagrams

(Example Problem 2: Graphical Method plus Write V and M Equations for Triangular Load)



$$x(52.8 - 9.26x^2)$$

$$x = \sqrt{\frac{52.8}{9.26}} = 7.55$$

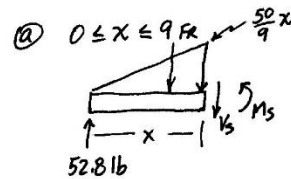
Support Reactions

$$+\circlearrowleft \sum M_A = 0; 225(6) + 200 - B_y(9) = 0$$

$$B_y = 172.2 \text{ lb}$$

$$+\uparrow \sum F_y = 0; A_y - 225 + 172 = 0; A_y = 52.8 \text{ lb}$$

$$A_x = 0$$

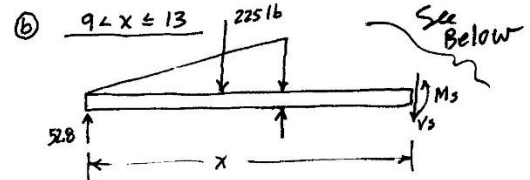


$$+\uparrow \sum F_y = 0; 52.8 - \left(\frac{50}{9}x\right)\left(x\right)\left(\frac{1}{2}\right) - V = 0;$$

$$\text{Root: } V = 0 \text{ at } x = 4.36' \Rightarrow V = 52.8 - 2.78x^2 \text{ lb}$$

$$+\circlearrowleft \sum M = 0; M - 52.8(x) + \left(\frac{50}{9}x\right)\left(x\right)\left(\frac{1}{2}\right)\left(\frac{1}{3}x\right) = 0$$

$$M = 52.8x - 0.926x^3 \text{ lb-ft}$$

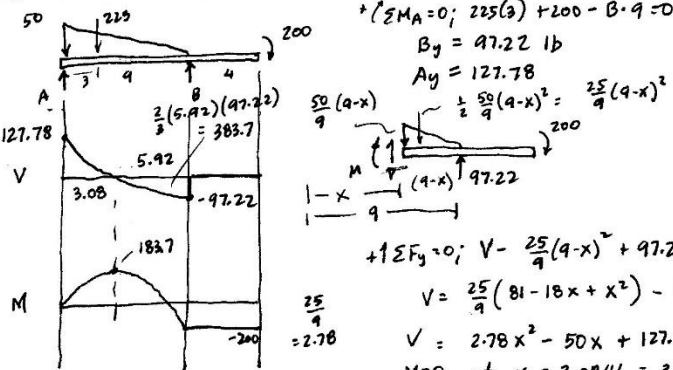


$$+\uparrow \sum F_y = 0; 52.8 - 225 + 172 - V = 0; V = 0$$

$$+\circlearrowleft \sum M = 0; -52.8(x) + 225(x-6) - 172(x-9) + M = 0$$

$$M = 200 \text{ lb-ft}$$

What if load reversed?



$$+\circlearrowleft \sum M_A = 0; 225(3) + 200 - B_y(9) = 0$$

$$B_y = 97.22 \text{ lb}$$

$$A_y = 127.78$$

$$\frac{50}{9}(9-x)$$

$$\frac{1}{2} \frac{50}{9}(9-x)^2 = \frac{25}{9}(9-x)^2$$

$$1-x$$

$$9-x$$

$$+\uparrow \sum F_y = 0; V - \frac{25}{9}(9-x)^2 + 97.22 = 0$$

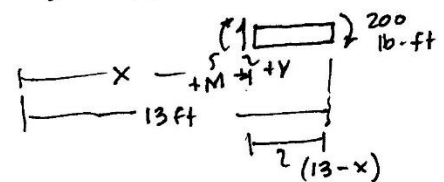
$$V = \frac{25}{9}(81 - 18x + x^2) - 97.22$$

$$V = 2.78x^2 - 50x + 127.78$$

$$V = 0 \text{ at } x = 3.0846 = 3.085$$

Much Easier:

Use free end...



$$+\uparrow V = 0$$

$$+\circlearrowleft M + 200 = 0;$$

$$M = -200 \text{ lb-ft}$$